MOHAN BABU UNIVERSITY

Sample Academic Assignment

Subject: LRRE

Student Name: _____
Student ID: _____
Submission Date: _____

Objective

Understand simple and multiple linear regression, model fitting, and evaluation.

Dataset

Use a suitable numeric dataset such as house prices, advertising data, or student study hours and marks.

Tasks

1. Identify the dependent variable and independent variable(s).

2. Plot the relationship between one independent variable and the dependent variable.

3. Fit a simple linear regression model and write its equation.

4. Interpret the slope and intercept in the context of the dataset.

5. Split the data into training and testing sets and fit a multiple linear regression model if multiple features are available.

6. Evaluate the model using MAE and R^2. Briefly explain what each metric indicates.

7. Mention two limitations or assumptions of linear regression.

Submission

Submit code/notebook, model equation, evaluation results, and a one-page interpretation.

Note: This is a sample practice assignment created for website testing and learning. It is not an official university-issued assignment.